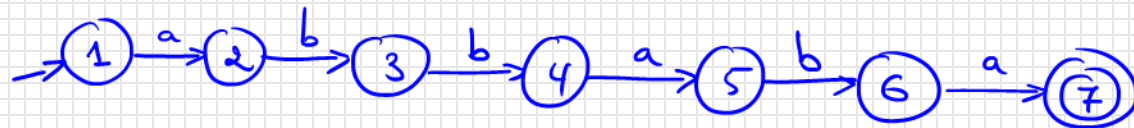


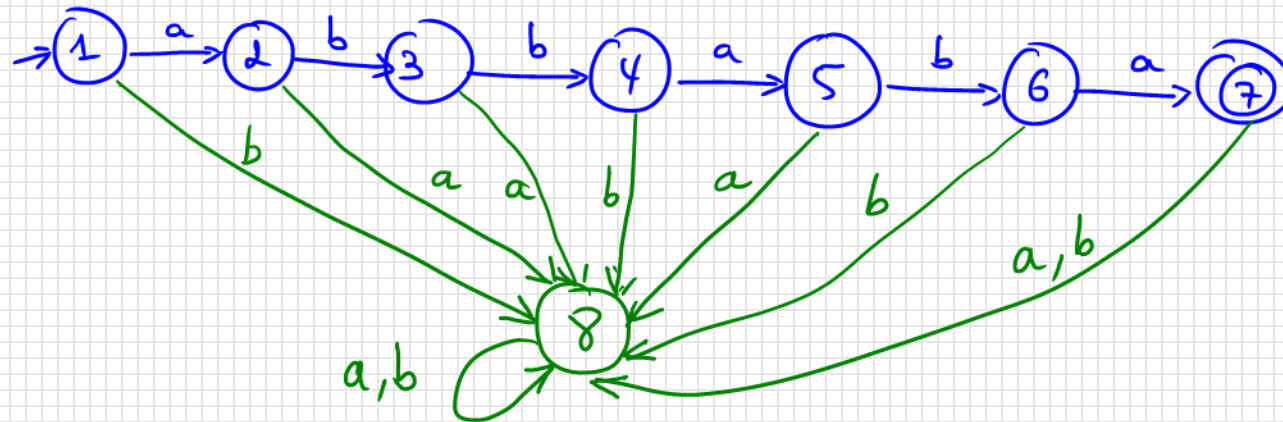
Automate Complémentaire

Donner un automate acceptant tous les mots sur $\Sigma = \{a, b\}$ sauf le mot abbaba?

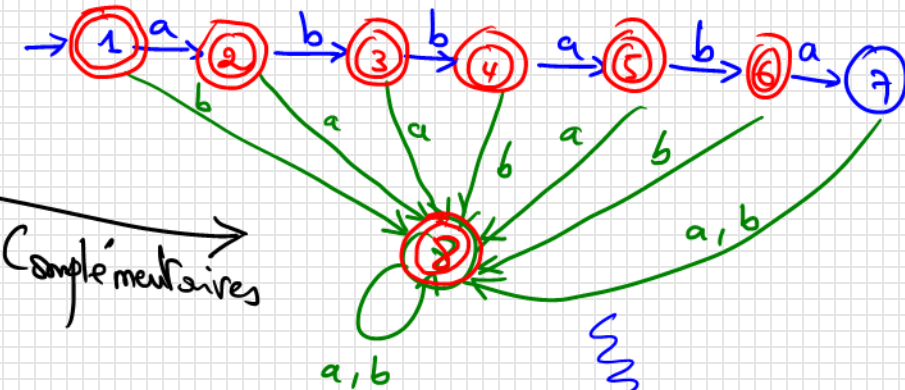
① Représenter l'automate acceptant le mot abbaba



② Rendre l'automate Complet



③ Inverser les Etats Finaux et les Etats Non Finaux.



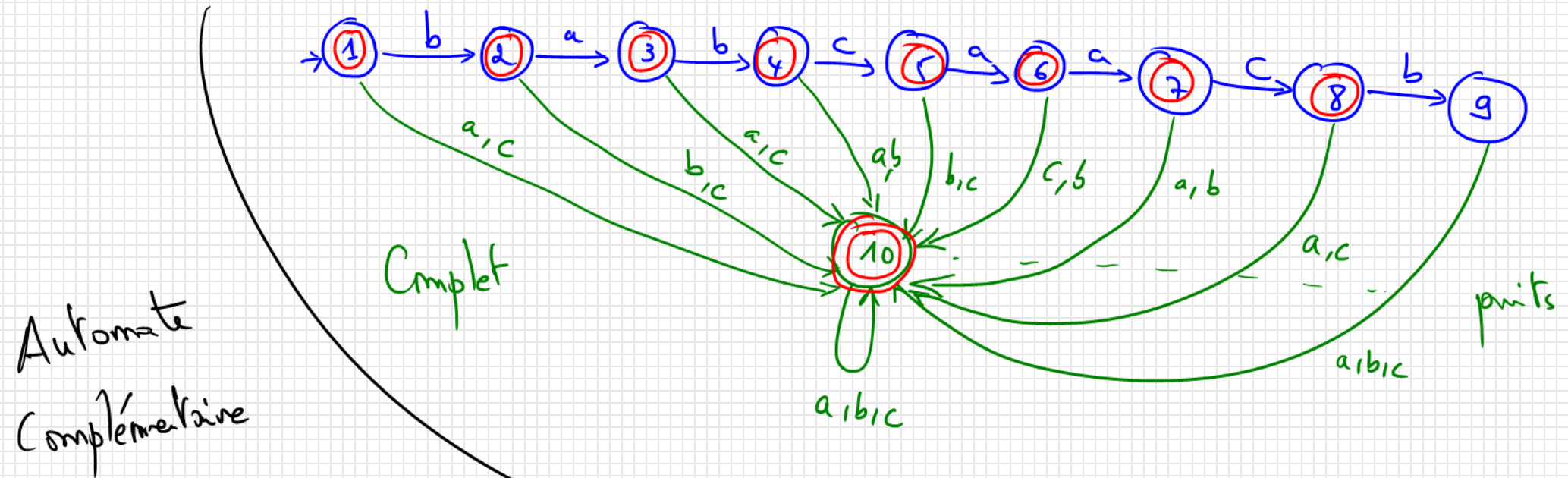
Deux Automates

Complémentaires

Automate Complémentaire

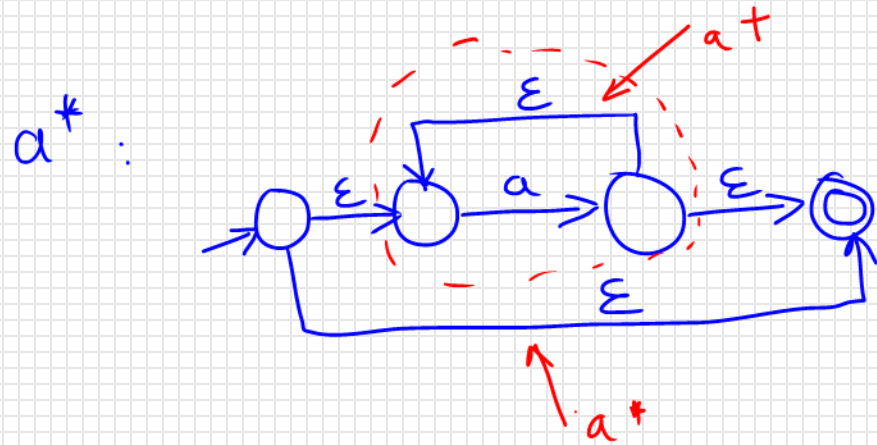
Application : $\Sigma = \{a, b, c\}$

Automate acceptant tous les mots sauf le mot `babcaacb` ?

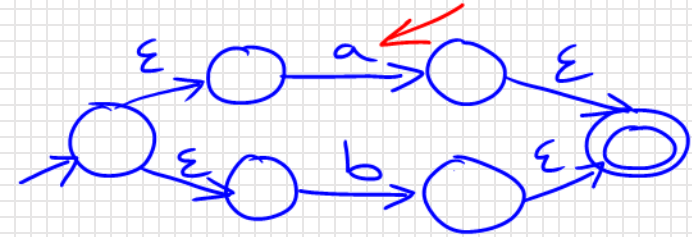


Algorithme de Thompson

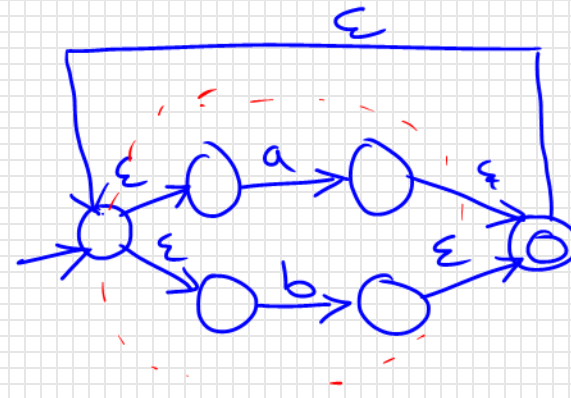
Tracer l'automate correspondant à n'importe quelle E.R ?



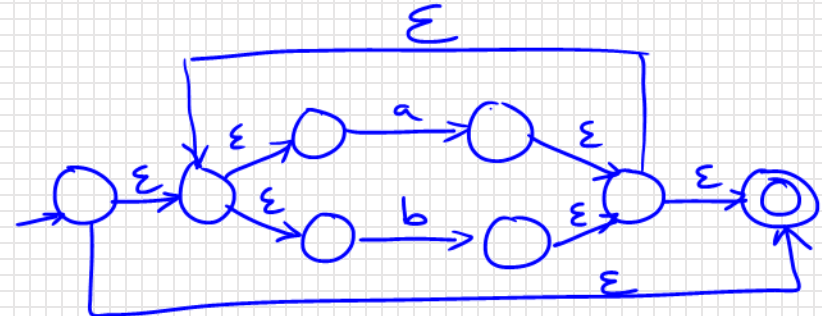
a/b :

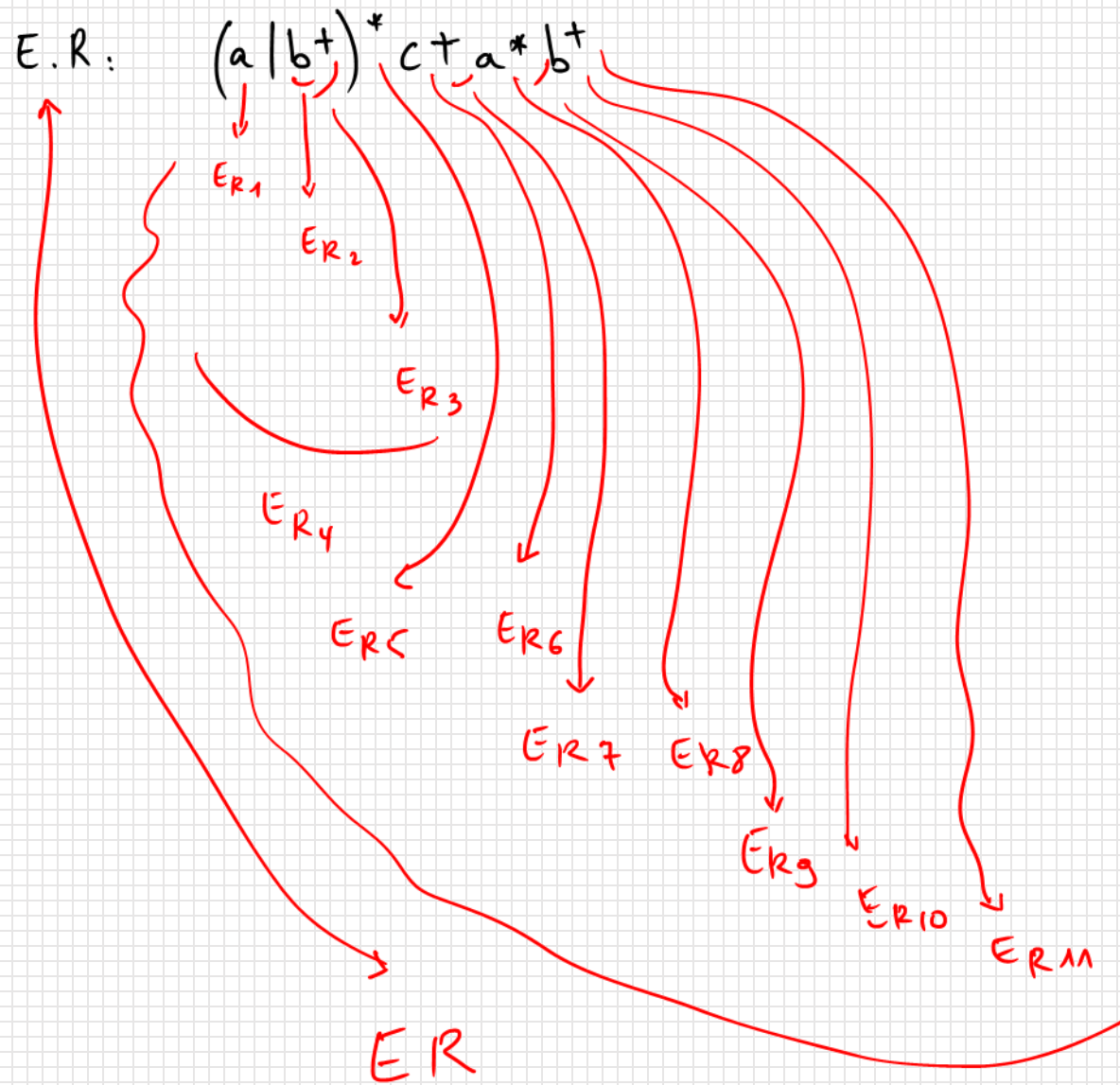


$(a/b)^+$:



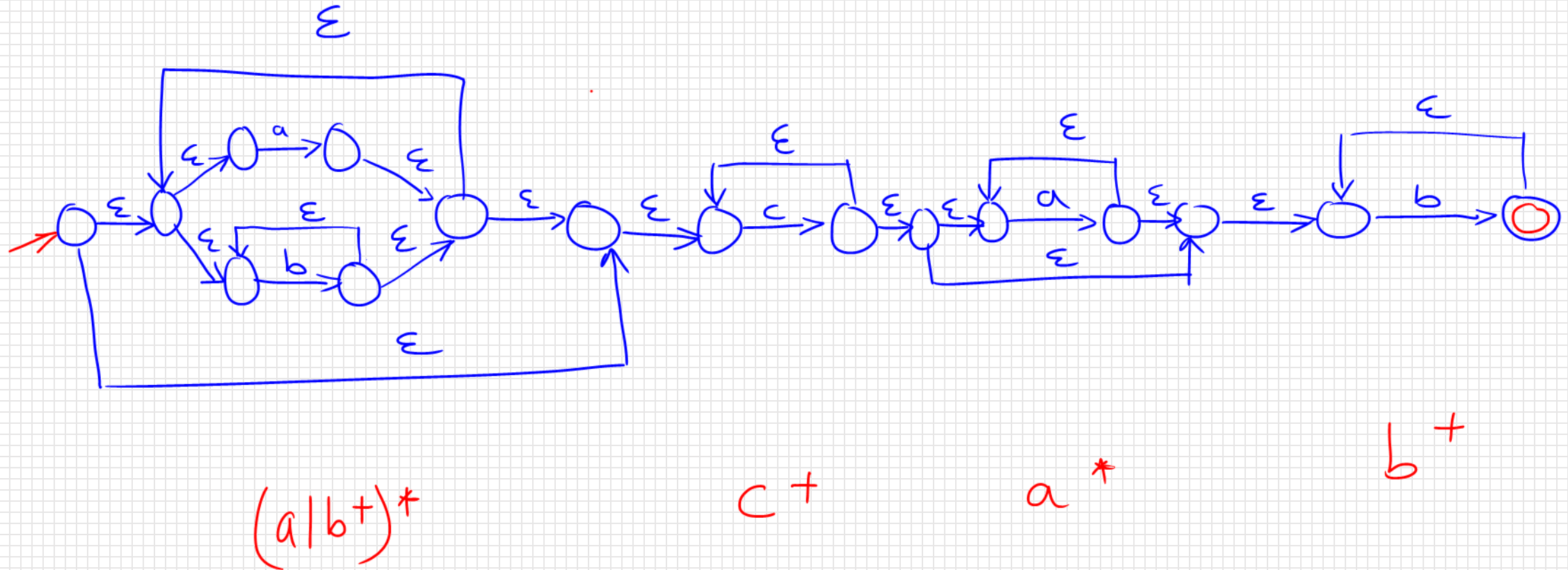
$(a/b)^*$:

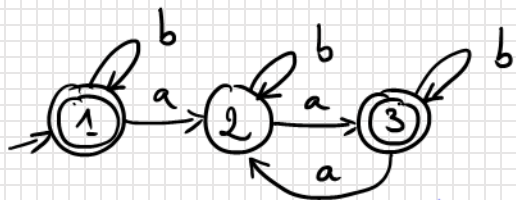




$$(a|b^+)^*c^+a^+b^+$$

Selon l'algorithme de Thompson

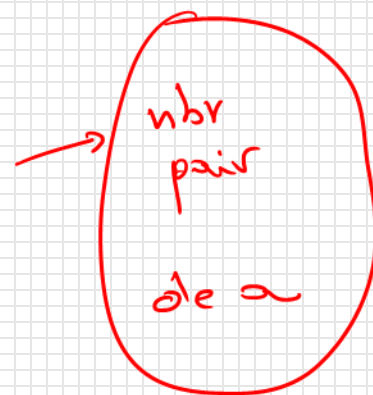




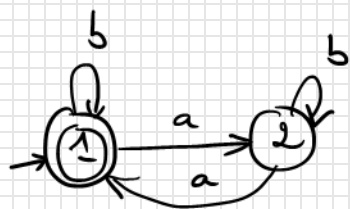
3 Etats

≡

ϵ
 a a
 b a b a
 a a b a b a
 ...



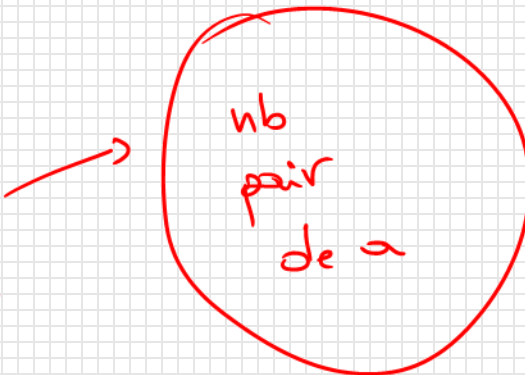
Minimisation



2 Etats

Automate Minimal

ϵ
 a a
 b
 a b a
 b a b a
 ...



automate
optimisé

Algorithme de Minimisation

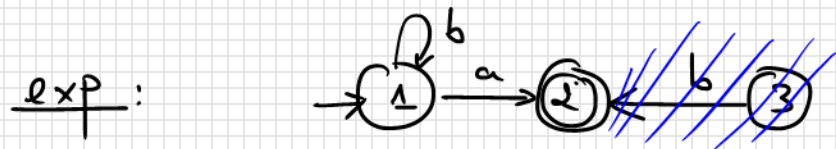
Comment obtenir l'automate minimal (Minimisation)

Avant de commencer la minimisation :

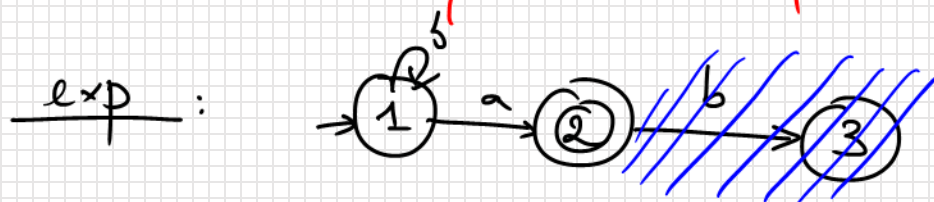
Pré-Traitement

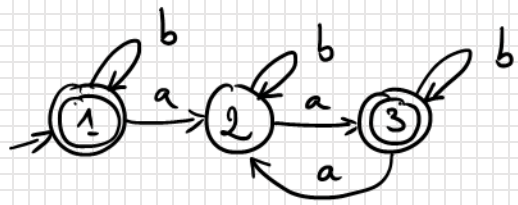
* il faut avoir un A.F.D (si non, le rendre déterministe)

* Eliminer les États Inaccessibles



* Eliminer les États qui ne mènent pas à un état final





A.F.D. ? ✓

E. Inacc. ? ✓

E. $\rightarrow$ E.F. ? ✓

① Regrouper les Etats en deux classes:

Etats Finaux $\rightarrow$ classe

Etats Non Finaux $\rightarrow$ autre classe

$I = \{1, 3\}$

$II = \{2\}$

② Tracer la Table de minimisation:

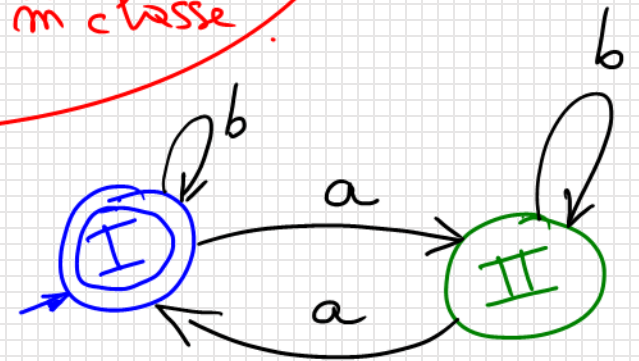
	1	3	2
Init.	I	I	II
a	II	II	I
b	I	I	II
Res	I	I	II

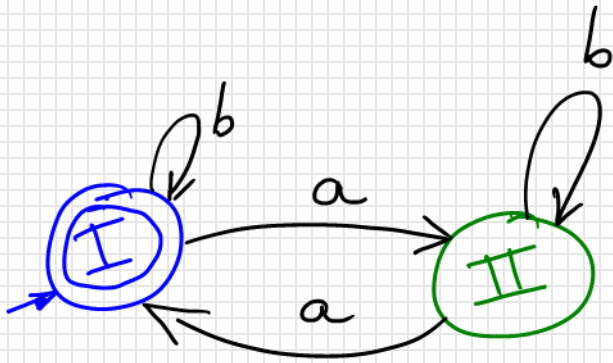
$\equiv$

Arrêt du Traitement

③ Comparer les Etats d'une m^{me} classe.

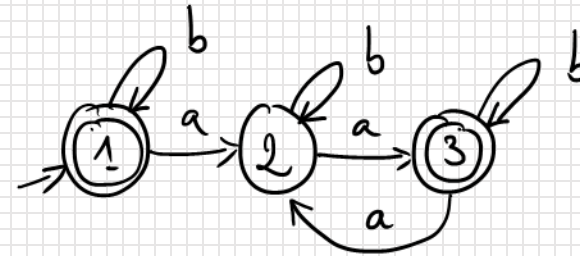
④ Tracer l'automate minimal:





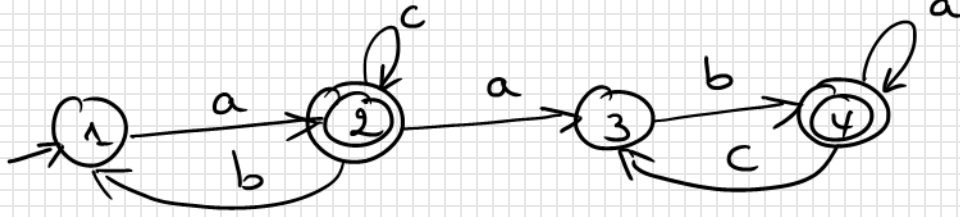
Automate Minimal

≡



Automate non minimal

$\Sigma = \{a, b, c\}$



Est-il minimal?

AFD? ✓

E. Inacc ✓

E. $\times \rightarrow$ EF ✓

$I = \{1, 3\}$: Non Finaux

$II = \{2, 4\}$: Finaux

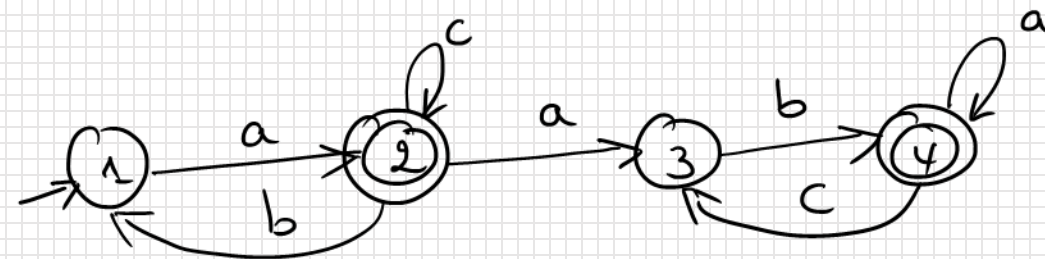
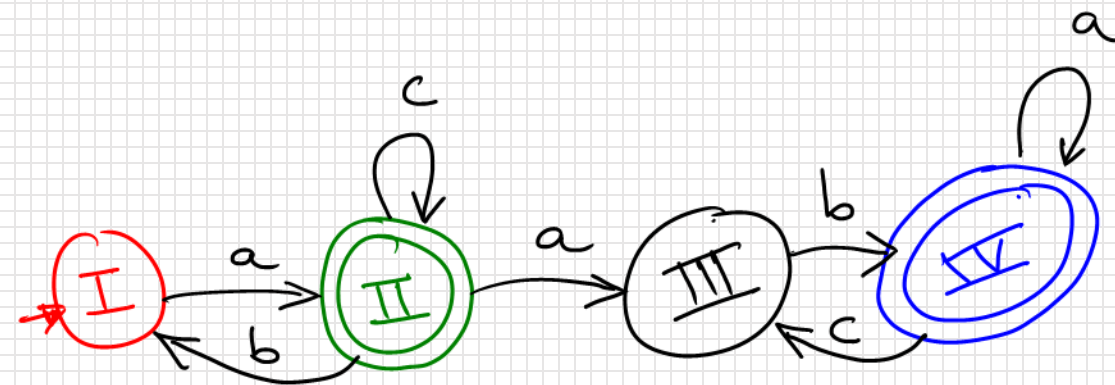
	1	3	2	4
Init	I	I	II	II
a	II	-	I	II
b	-	II	I	-
c	-	-	II	I
Res1	I	III	II	IV
a	II	-	III	IV
b	-	IV	I	-
c	-	-	II	III
Res2	I	III	II	IV

$\neq ?$



Arrêt du Traitement

	1	3	2	4
Init	I	I	II	II
a	II	-	I	II
b	-	II	I	-
c	-	-	II	I
Res1	I	III	II	IV
a	II	-	III	IV
b	-	IV	I	-
c	-	-	II	III
Res2	I	III	II	IV



Oui, il est minimal.